

Math in Python

Python can be used to solve all sorts of mathematical problems, including addition, subtraction, multiplication, and division. Variables can also be used in sums.

SEE ALSO

◀ 52–53 Math

◀ 108–109 Variables in Python

Simple calculations

In Python, simple calculations can be made by typing them into the shell window. The “print()” function is not needed for this—Python gives the answer straight away. Try these examples in the shell window:

```
>>> 12 + 4
16
```

△ Addition

Use the “+” symbol to add numbers together.

Use the shell window to get instant results

The answer appears when you press “Enter”

```
>>> 12 - 4
8
```

△ Subtraction

Use the “-” symbol to subtract the second number from the first one.

You can't divide by zero, so you'll always get an error if you try to do so.



```
>>> 12 * 4
48
```

△ Multiplication

Use the “*” symbol to multiply two numbers together.

Computers use the “*” symbol, not “x”, for multiplication

```
>>> 12 / 4
3.0
```

△ Division

Use the “/” symbol to divide the first number by the second one.

Division in Python gives an answer as a float (a number with a decimal point)

Using brackets

Brackets can be used to instruct Python which part of a sum to do first. Python will always work out the value of the sum in the bracket, before solving the rest of the problem.

First it works out that $6 + 5 = 11$, then 11 is multiplied by 3

```
>>> (6 + 5) * 3
33
```

△ Addition first

In this sum, brackets are used to instruct Python to do the addition first.

First it works out that $5 * 3 = 15$, then 15 is added to 6

```
>>> 6 + (5 * 3)
21
```

Different answer

△ Multiplication first

Brackets here are used to do the multiplication first, in order to end up with the correct answer.